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# Dispatchable Virtual Oscillator Control (dVOC) for a 2L-VSI

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This model builds a three-phase *grid-forming* inverter whose outer control loop is not the usual droop controller but a nonlinear *oscillator*, the Andronov–Hopf oscillator, the heart of *dispatchable Virtual Oscillator Control* (dVOC). Everything is built from plain PLECS blocks (no C-code). Background comes from (Rosso et al., 2021) and (Seo et al., 2019)

## 1 The big picture: why an oscillator?

Most inverters are *grid-following*: a PLL reads the grid angle and the inverter provides current behind that angle. That only works if the voltage is being held up (a stiff grid is required), if that's not the case, then we need inverters that can *be* the grid instead of just following it. Those are *grid-forming* (GFM) inverters: they behave like a controllable AC voltage source and let the power sort itself out from the physics of the line.

Droop control is the classic way to do this. *Virtual oscillator control* (VOC) is a newer idea: instead of computing an angle, you let a nonlinear oscillator *be* the AC waveform — its stable, self-sustaining oscillation is the output voltage.

**The one idea to keep in mind.** A three-phase voltage is, at bottom, a **rotating vector**: a two-dimensional arrow  $\mathbf{v} = [v_\alpha, v_\beta]^\top$  in the stationary  $\alpha\beta$  plane. Everything you care about is contained in how that arrow moves:

- its **length**  $\|\mathbf{v}\|$  is the voltage *magnitude*;
- its **angular speed**  $d\theta/dt$  is the *frequency*;
- its **angle** relative to the grid's own arrow is what sets the *power flow*.

So a grid-forming controller has exactly one job: *drive a vector around a circle, and lean on its length and its speed when the power is wrong*. That is the whole of dVOC, and the rest of this document is just writing that sentence as a differential equation.

The way we get the vector to spin on its own is with a virtual oscillator: a two-state dynamical system, written in software, whose natural behaviour already *is* a rotation. We then bolt on two corrections — one that fixes the length, one that fixes the speed and the length according to the power error. The nice result is that a bunch of these oscillators automatically lock to the same frequency and voltage, and in steady state the whole thing behaves *exactly* like droop — but unlike droop it stays stable during big transients and black starts. The *dispatchable* version (dVOC) just adds power set-points  $P^*, Q^*$  so you can command it where to go. That is what we built.

## 2 The oscillator core

### 2.1 Step 1: the skeleton: an ideal LC tank

We need an object that rotates on its own. The simplest one in electrical engineering is a lossless *LC* tank: energy sloshes back and forth between the capacitor and the inductor forever. Its capacitor

voltage  $v_c$  and inductor current  $i_L$  obey

$$\boxed{\begin{aligned} L \frac{di_L}{dt} &= v_c, \\ C \frac{dv_c}{dt} &= -i_L, \end{aligned}} \quad (1)$$

which is a sustained oscillation at the natural frequency

$$\omega_0 = \frac{1}{\sqrt{LC}}. \quad (2)$$

Two remarks worth carrying forward. First, this tank is **virtual**:  $L$  and  $C$  are numbers inside the control code, not components on a board, so we are free to choose them and we will choose them so that  $\omega_0$  lands exactly on the nominal grid frequency. Second, (1) is only *marginally* stable: it rings at  $\omega_0$  forever, but at **whatever amplitude the initial condition happened to hand it**, and there is no input anywhere to tell it anything. A useful controller has to fix both of those. That is precisely what the extra terms below do.

## 2.2 Step 2: the four extra terms

dVOC keeps the  $LC$  skeleton and adds two virtual sources plus two external forcing signals:

$$\boxed{\begin{aligned} L \frac{di_L}{dt} &= v_c + v_m - \varepsilon u_2, \\ C \frac{dv_c}{dt} &= -i_L + i_m - u_1. \end{aligned}} \quad (3)$$

Reading (3) one symbol at a time:

| Symbol     | What it is                                                                  | What it does                                                                                                                                                                                                                                                                               |
|------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $v_c, i_L$ | the two <b>states</b> of the virtual circuit                                | The oscillator itself. Together they are the rotating vector                                                                                                                                                                                                                               |
| $v_m, i_m$ | virtual <b>voltage</b> and <b>current source</b> , for amplitude correction | A dependent source in series with the inductor and one in parallel with the capacitor. They inject or absorb energy so the oscillation settles on one specific amplitude instead of drifting. This is what turns marginal stability into a <i>stable limit cycle</i> .                     |
| $u_1, u_2$ | <b>external forcing</b>                                                     | The only place the outside world enters. They carry the current error (how far the converter is from delivering the commanded $P^*, Q^*$ ) back into the oscillator, and thereby steer its angle and its size. Fully developed in Section 3; for now just note <i>that</i> they are there. |

The  $\varepsilon$  multiplying  $u_2$  looks arbitrary at this point. It is not: it is exactly what makes the two forcing channels enter the vector equation with the same gain, as the next two steps show.

### 2.3 Step 3: choosing the states so the trajectory is a circle

$v_c$  is a voltage and  $i_L$  is a current, so plotting one against the other gives an *ellipse*, not a circle — and an ellipse is useless as a phasor, because its length would change as it spins. The fix is to rescale the current by the tank's characteristic impedance,

$$\varepsilon \triangleq \sqrt{\frac{L}{C}} \quad [\Omega], \quad (4)$$

so that  $\varepsilon i_L$  carries units of volts and the same peak value as  $v_c$ . That gives the state vector

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} v_c \\ \varepsilon i_L \end{bmatrix}, \quad \|\mathbf{x}\|^2 = v_c^2 + \varepsilon^2 i_L^2. \quad (5)$$

Now  $\|\mathbf{x}\|$  is genuinely the *length of the phasor*, constant on a circular orbit. It is the single scalar the amplitude loop will regulate.

Two identities follow straight from (4) and (2), and they are the only algebra needed in Step 4:

$$\frac{1}{C} = \omega_0 \varepsilon, \quad \frac{1}{L} = \frac{\omega_0}{\varepsilon}. \quad (6)$$

**What  $v_m$  and  $i_m$  actually are.** They are the amplitude correction, defined as

$$v_m = \frac{\xi}{\omega_0} g(\mathbf{x}) \varepsilon i_L, \quad (7)$$

$$i_m = \frac{\xi}{\varepsilon \omega_0} g(\mathbf{x}) v_c, \quad (8)$$

where

$$g(\mathbf{x}) \triangleq v^{*2} - \|\mathbf{x}\|^2, \quad v^{*2} \triangleq 2 \left( \frac{V_{\text{ref}}}{k_v} \right)^2. \quad (9)$$

$g$  is simply the **amplitude error**: how far the squared length of the phasor sits from its target  $v^{*2}$ . Notice that each source is proportional to *its own* state ( $v_m$  to  $\varepsilon i_L$ ,  $i_m$  to  $v_c$ ) and that the  $\varepsilon$  and  $\omega_0$  factors are placed exactly so that both collapse to the same expression in Step 4.

### 2.4 Step 4: rewriting it as one vector ODE

Differentiate (5) and substitute (3), using the identities (6). For the first state,

$$\dot{x}_1 = \frac{dv_c}{dt} = \frac{1}{C} (-i_L + i_m - u_1) = \omega_0 \varepsilon (-i_L + i_m - u_1) = -\omega_0 \underbrace{\varepsilon i_L}_{x_2} + \omega_0 \varepsilon i_m - \omega_0 \varepsilon u_1, \quad (10)$$

and for the second,

$$\dot{x}_2 = \varepsilon \frac{di_L}{dt} = \frac{\varepsilon}{L} (v_c + v_m - \varepsilon u_2) = \omega_0 (v_c + v_m - \varepsilon u_2) = +\omega_0 \underbrace{v_c}_{x_1} + \omega_0 v_m - \omega_0 \varepsilon u_2. \quad (11)$$

Both forcing terms come out with the *same* gain  $\omega_0 \varepsilon$ , that is what the  $\varepsilon$  next to  $u_2$  in (3) was for. Now insert the amplitude sources (7), (8):

$$\omega_0 \varepsilon i_m = \omega_0 \varepsilon \cdot \frac{\xi}{\varepsilon \omega_0} g v_c = \xi g x_1, \quad \omega_0 v_m = \omega_0 \cdot \frac{\xi}{\omega_0} g \varepsilon i_L = \xi g x_2. \quad (12)$$

Both reduce to  $\xi g$  times *their own state*. The two scalar equations are therefore

$$\begin{aligned}\dot{x}_1 &= -\omega_0 x_2 + \xi g(\mathbf{x}) x_1 - \omega_0 \varepsilon u_1, \\ \dot{x}_2 &= +\omega_0 x_1 + \xi g(\mathbf{x}) x_2 - \omega_0 \varepsilon u_2,\end{aligned}\tag{13}$$

which is the same equation twice, once per axis, so it collapses into a single vector equation. The last step is to name the output. The  $\alpha\beta$  voltage the modulator receives is the state scaled by the internal voltage gain  $k_v$ :

$$\mathbf{v} = \begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} = k_v \mathbf{x} \quad \implies \quad v_\alpha = k_v v_c, \quad v_\beta = k_v \varepsilon i_L.\tag{14}$$

We use  $k_v = 1$ , so  $\mathbf{v} = \mathbf{x}$  and everything can be written directly in terms of the voltage vector:

$$\dot{\mathbf{v}} = \underbrace{\omega_0 \mathbf{J} \mathbf{v}}_{\text{1. oscillation}} + \underbrace{\xi (v^{*2} - \|\mathbf{v}\|^2) \mathbf{v}}_{\text{2. amplitude correction}} - \underbrace{\omega_0 \varepsilon \mathbf{u}}_{\text{3. power adaptation}}, \quad \mathbf{J} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.\tag{15}$$

This is the whole controller. Everything that follows is just unpacking these three terms.

## 2.5 Reading the three terms

1. **Oscillation.**  $\omega_0 \mathbf{J} \mathbf{v}$  spins the vector at constant angular speed  $\omega_0$ .  $\mathbf{J}$  is a  $90^\circ$  rotation, so this term is always perpendicular to  $\mathbf{v}$ : it changes the *angle* and never the *length*. On its own it draws a perfect circle at the nominal frequency, it is the *LC* tank of Step 1.
2. **Amplitude correction.**  $\xi g(\mathbf{v}) \mathbf{v}$  is parallel to  $\mathbf{v}$ , so it does the opposite: it changes the *length* and never the *angle*. Its sign flips at  $\|\mathbf{v}\|^2 = v^{*2}$ , which pins the orbit to one radius. This is the term that makes the oscillator a *stable* oscillator rather than a marginal one.
3. **Frequency and amplitude adaptation from power.**  $-\omega_0 \varepsilon \mathbf{u}$  is the only term that knows anything about the grid. It carries the current error, and by pushing the vector partly along itself and partly across itself it *detunes* both the speed and the length until the measured power matches  $P^*, Q^*$ . This is where droop comes from and the split between “along” and “across” is set by a single angle  $\phi$ .

## 3 Unpacking the three terms

### 3.1 Term 1: oscillation, written out

Expanding  $\mathbf{J}$  against the  $\alpha\beta$  components:

$$\omega_0 \mathbf{J} \mathbf{v} = \omega_0 \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} = \omega_0 \begin{bmatrix} -v_\beta \\ v_\alpha \end{bmatrix},\tag{16}$$

i.e.  $\dot{v}_\alpha = -\omega_0 v_\beta$  and  $\dot{v}_\beta = +\omega_0 v_\alpha$ : the classic quadrature pair. Because  $\mathbf{J} \mathbf{v}$  is orthogonal to  $\mathbf{v}$  (so  $\mathbf{v}^\top \mathbf{J} \mathbf{v} = 0$ ), this term contributes nothing to  $\frac{d}{dt} \|\mathbf{v}\|^2$ . With terms 2 and 3 switched off, (15) integrates to

$$\mathbf{v}(t) = \|\mathbf{v}(0)\| \begin{bmatrix} \cos(\omega_0 t + \theta_0) \\ \sin(\omega_0 t + \theta_0) \end{bmatrix},\tag{17}$$

which an inverse Clarke transform turns into a balanced three-phase set for the PWM — with no angle ever computed explicitly. *The oscillator is the angle generator.*

### 3.2 Term 2: amplitude correction, and the constants

Written out, term 2 is

$$\xi g(\mathbf{v}) \mathbf{v} = \xi \left( v^{*2} - v_\alpha^2 - v_\beta^2 \right) \begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix}, \quad v^{*2} = 2 \left( \frac{V_{\text{ref}}}{k_v} \right)^2 \stackrel{k_v=1}{=} 2V_{\text{ref}}^2. \quad (18)$$

The bracket is the clever bit, a damping term that changes sign:

- Oscillation *too small* ( $\|\mathbf{v}\|^2 < v^{*2}$ ): bracket positive  $\Rightarrow$  it *adds* energy, the amplitude grows.
- Oscillation *too big* ( $\|\mathbf{v}\|^2 > v^{*2}$ ): bracket negative  $\Rightarrow$  it *removes* energy, the amplitude shrinks.
- Exactly right ( $\|\mathbf{v}\|^2 = v^{*2}$ ): nothing happens, a stable *limit cycle*.

So from almost any starting point the oscillation spirals out (or in) onto a circle of radius  $\sqrt{v^{*2}} = \sqrt{2} V_{\text{ref}}$  and stays there. (This self-correcting behaviour is a supercritical *Andronov–Hopf bifurcation*, hence the name.) The knob  $\xi$  (xi) sets *how fast* it settles; linearising  $\|\mathbf{v}\|^2$  about  $v^{*2}$  gives

$$\tau_{\text{amp}} \approx \frac{1}{4\xi V_{\text{ref}}^2}. \quad (19)$$

### 3.3 Term 3a: from power set-points to a reference current

The oscillator sets the voltage; the third term needs a *current* to compare against. So we ask: given the voltage the oscillator is producing right now, what current would deliver exactly  $P^*$  and  $Q^*$ ? Instantaneous power in the  $\alpha\beta$  frame (amplitude-invariant Clarke) is

$$P = \frac{3}{2}(v_\alpha i_\alpha + v_\beta i_\beta), \quad Q = \frac{3}{2}(v_\beta i_\alpha - v_\alpha i_\beta), \quad (20)$$

which stacks into

$$\begin{bmatrix} P \\ Q \end{bmatrix} = \frac{3}{2} \underbrace{\begin{bmatrix} v_\alpha & v_\beta \\ v_\beta & -v_\alpha \end{bmatrix}}_{\mathbf{M}} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix}. \quad (21)$$

The point of writing it this way is that  $\mathbf{M}$  is symmetric with  $\mathbf{M}^2 = \|\mathbf{v}\|^2 \mathbf{I}$ , so its inverse is just  $\mathbf{M}/\|\mathbf{v}\|^2$ , no numerical matrix inversion is needed anywhere. Solving (21) for the current that would deliver the commands gives the dispatch matrix  $\Gamma$ :

$$\boxed{\mathbf{i}^* = \begin{bmatrix} i_\alpha^* \\ i_\beta^* \end{bmatrix} = \underbrace{\frac{2}{3\|\mathbf{v}\|^2} \begin{bmatrix} v_\alpha & v_\beta \\ v_\beta & -v_\alpha \end{bmatrix}}_{\Gamma} \begin{bmatrix} P^* \\ Q^* \end{bmatrix}, \quad \|\mathbf{v}\|^2 = v_\alpha^2 + v_\beta^2.} \quad (22)$$

Component-wise,

$$i_\alpha^* = \frac{2(P^* v_\alpha + Q^* v_\beta)}{3(v_\alpha^2 + v_\beta^2)}, \quad i_\beta^* = \frac{2(P^* v_\beta - Q^* v_\alpha)}{3(v_\alpha^2 + v_\beta^2)}. \quad (23)$$

### 3.4 Term 3b: the rotated current error

Now compare the reference (22) against the measured grid-side current  $\mathbf{i}_o = [i_\alpha, i_\beta]^\top$  (taken at the PCC, after the LCL filter), scale it by the sync gain  $k_i$ , and rotate the result by an angle  $\phi$ :

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = k_i \mathbf{R}(\phi) (\mathbf{i}_o - \mathbf{i}^*) = \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix} \begin{bmatrix} k_i (i_\alpha - i_\alpha^*) \\ k_i (i_\beta - i_\beta^*) \end{bmatrix}. \quad (24)$$

Expanded channel by channel,

$$\begin{aligned} u_1 &= k_i \left[ \cos \phi (i_\alpha - i_\alpha^*) - \sin \phi (i_\beta - i_\beta^*) \right], \\ u_2 &= k_i \left[ \sin \phi (i_\alpha - i_\alpha^*) + \cos \phi (i_\beta - i_\beta^*) \right]. \end{aligned} \quad (25)$$

Both channels carry the *same* gain  $k_i$ : the rotation is applied once, to the error vector as a whole. Substituting (24) into (15) and folding the gain  $\omega_0 \varepsilon$  into  $\eta \triangleq \omega_0 \varepsilon k_i$ , the minus sign flips the error around and term 3 reads

$$-\omega_0 \varepsilon \mathbf{u} = \eta \mathbf{R}(\phi) (\mathbf{i}^* - \mathbf{i}_o), \quad (26)$$

i.e. “if you are delivering less current than commanded, push the vector in the  $\mathbf{R}(\phi)$  direction.” Because  $\mathbf{R}(\phi)$  mixes the radial and tangential directions, that push lands partly on the **length** of  $\mathbf{v}$  (a voltage correction, which moves  $Q$ ) and partly on its **angle** (a frequency correction, which moves  $P$ ).  $\phi$  is the dial that sets the mix and that mix is precisely how a converter with no droop curve programmed into it ends up behaving exactly like one.

**What  $\phi$  does.** It matches the line’s impedance angle. For a mostly *inductive* line,  $\phi = \pi/2$  cleanly separates “ $P$  moves the angle” from “ $Q$  moves the voltage” (the familiar droop split). We ship  $\phi = \pi/2$  in our application. It is the paper’s  $\kappa$ .

**What  $k_i$  does.** It’s the sync gain: bigger  $k_i$  chases the power reference faster but reacts harder to grid disturbances.

### 3.5 Putting it all together

Collecting (16), (18) and (26) into (15) gives the complete dVOC law in vector form:

$$\dot{\mathbf{v}} = \omega_0 \mathbf{J} \mathbf{v} + \xi (v^{*2} - \|\mathbf{v}\|^2) \mathbf{v} + \eta \mathbf{R}(\phi) \left[ \frac{2}{3\|\mathbf{v}\|^2} \begin{bmatrix} v_\alpha & v_\beta \\ v_\beta & -v_\alpha \end{bmatrix} \begin{bmatrix} P^* \\ Q^* \end{bmatrix} - \mathbf{i}_o \right], \quad (27)$$

with  $\eta = \omega_0 \varepsilon k_i$  and  $v^{*2} = 2V_{\text{ref}}^2$ . Written out per axis, with every matrix expanded, this is exactly what the PLECS block diagram computes:

$$\begin{aligned} \dot{v}_\alpha &= -\omega_0 v_\beta + \xi (2V_{\text{ref}}^2 - v_\alpha^2 - v_\beta^2) v_\alpha + \eta \left[ \cos \phi (i_\alpha^* - i_\alpha) - \sin \phi (i_\beta^* - i_\beta) \right], \\ \dot{v}_\beta &= +\omega_0 v_\alpha + \xi (2V_{\text{ref}}^2 - v_\alpha^2 - v_\beta^2) v_\beta + \eta \left[ \sin \phi (i_\alpha^* - i_\alpha) + \cos \phi (i_\beta^* - i_\beta) \right], \end{aligned} \quad (28)$$

$$i_\alpha^* = \frac{2(P^* v_\alpha + Q^* v_\beta)}{3(v_\alpha^2 + v_\beta^2)}, \quad i_\beta^* = \frac{2(P^* v_\beta - Q^* v_\alpha)}{3(v_\alpha^2 + v_\beta^2)}. \quad (29)$$

**The shipped case,**  $\phi = \pi/2$ . Since  $\cos \phi = 0$  and  $\sin \phi = 1$ ,  $\mathbf{R}(\pi/2) = \mathbf{J}$  and (28) collapses to

$$\begin{aligned} \dot{v}_\alpha &= \underbrace{-\omega_0 v_\beta}_{1. \text{ oscillation}} + \underbrace{\xi(2V_{\text{ref}}^2 - \|\mathbf{v}\|^2)v_\alpha}_{2. \text{ amplitude correction}} - \underbrace{\eta(i_\beta^* - i_\beta)}_{3. \text{ power adaptation}}, \\ \dot{v}_\beta &= \underbrace{+\omega_0 v_\alpha}_{1. \text{ oscillation}} + \underbrace{\xi(2V_{\text{ref}}^2 - \|\mathbf{v}\|^2)v_\beta}_{2. \text{ amplitude correction}} + \underbrace{\eta(i_\alpha^* - i_\alpha)}_{3. \text{ power adaptation}}. \end{aligned} \quad (30)$$

Each of the three sources of motion is now visible on its own line: a rotation, a radial pull towards the target amplitude, and a 90°-rotated current error. Nothing else runs in the outer loop.

## 4 Per-unit inside, real volts outside

The oscillator runs in per-unit and only gets scaled to real volts and amps at its terminals:

- **Tank.**  $k_v = 1$ ,  $C_{\text{osc}} = 1$ ,  $L_{\text{osc}} = 1/\omega_g^2$  so the natural frequency lands on the grid frequency. The scaling factor is the tank’s characteristic impedance  $\varepsilon = \sqrt{L_{\text{osc}}/C_{\text{osc}}} = 1/\omega_g$ , which is what keeps  $x = [v_c, \varepsilon i_L]^\top$  a nice circle.
- **Voltage out.** Since the peak is  $\sqrt{2}V_{\text{ref}}$ , the output gain  $K_{v,\text{out}} = V_{\text{base, FN, peak}}/\sqrt{2} = V_{\text{base, RMS}}/\sqrt{3}$  maps  $V_{\text{ref}} = 1$  to the rated RMS phase voltage ( $\approx 63.5$  kV on a 110 kV line-to-line base).
- **Current in.** Measured currents are divided by  $I_{\text{norm}} = I_{\text{base, peak}}$ , so  $k_i$  stays a per-unit gain.

## 5 Three add-ons you can toggle

These next concepts sit around the basic dVOC. Each is there to teach one idea, and each can be switched off so you can see the “before” and “after”.

### 5.1 Pre-synchronisation (soft grid connection)

Close the breaker while the oscillator is at a random phase and you get an inrush spike. So before connecting, we push a virtual “line-up” current into the error loop,

$$i_{\text{sync}} = K_{\text{sync}}(v_{\text{inv}} - v_{\text{grid}}), \quad K_{\text{sync}} = k_{\text{sync, gain}} \frac{I_{\text{norm}}}{V_{\text{base, FN, peak}}}, \quad (31)$$

which nudges the inverter voltage until it matches the grid, so at closure there’s almost no jump. Turn it on with `presync_en`; the window runs from `t_presync` to grid connection `t_gridon`. It cut the inrush here by about 25× ( $\sim 7.6 \rightarrow 0.30$  p.u.).

### 5.2 Virtual resistance (damping)

The steps we used for simulation, ring the lightly-damped LCL filter. Instead of a real (lossy) resistor, we just subtract a fake resistive drop from the modulating voltage:

$$v_{\text{pwm}} = v_{\text{dVOC}} - R_v i_{\text{PCC}}. \quad (32)$$

$R_v$  (`Rv_damp`) damps the ringing like a lossless output resistor. The catch is a small voltage droop, so keep it modest (a few % of  $Z_{\text{base}}$ ); set `Rv_damp = 0` to switch it off.

### 5.3 Integral $Q \rightarrow V$ loop (with anti-windup)

On its own, dVOC tracks  $Q$  with a *droop* offset, to nail  $Q$  exactly, an outer integrator trims the amplitude target:

$$V_{\text{ref}^2, \text{dyn}} = 2V_{\text{ref}}^2 + q_{\text{en}} \int K_{qv} (Q^* - Q) dt. \quad (33)$$

Turn it on with `q_en`.

#### What to watch out for

- **No built-in current limit.** As a voltage source it won't cap its own current; faults/large steps need a separate limiter (not in this model yet).
- **$Q$  needs help:** exact reactive tracking needs the extra integral loop, which needs anti-windup to stay safe.
- **Depends on  $\phi$ :** if  $\phi$  doesn't match the line angle,  $P$  and  $Q$  start to interfere each other.
- **LCL ringing:** the filter is lightly damped; virtual resistance helps but doesn't fully cure it.

## 6 Running it: the built-in test sequence

### References

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